

Statistical & Measurement Principles in GradeCast

Summary Hierarchy

Domain	Core Formulas	Purpose
ANN Regression	MSE, Adam, R ²	Predict unseen item scores
IRT (PCM/RSM)	Logistic probability functions	Calibrate item/person parameters
Reliability	$\frac{Var(\theta)}{Var(\theta) + Var_{error}}$	Internal consistency of latent scale
Fit Stats	Infit, Outfit	Item performance quality
Validation	Holdout / LOOCV	Generalization assessment
Visualization	Loss, Scatter, Wright Map	Interpretability and transparency

1. Descriptive Statistics

a. Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Average of observations; measures central tendency.

b. Variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Measures data dispersion around the mean.

c. Standard Deviation

$$s = \sqrt{s^2}$$

Indicates spread of values in same units as data.

d. Covariance and Correlation

$$Cov(X, Y) = \frac{1}{n-1} \sum_i (x_i - \bar{x})(y_i - \bar{y})$$
$$r = \frac{Cov(X, Y)}{s_X s_Y}$$

Shows strength and direction of linear association.

2. Artificial Neural Network (ANN) Regression

a. Model Architecture

$$\hat{y} = f(X; W, b) = \sigma(W_2 \cdot \sigma(W_1 X + b_1) + b_2)$$

ANNs approximate nonlinear relationships between features and student scores.

b. Activation Functions

- **ReLU:** $f(x) = \max(0, x)$
- **Sigmoid:** $f(x) = \frac{1}{1+e^{-x}}$
- **Tanh:** $f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

c. Loss Function (Mean Squared Error)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Minimized during training to improve prediction accuracy.

d. Optimization (Adam)

Gradient-based learning using moment estimates:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) L_t \nabla_{\theta},$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (L_t \nabla_{\theta})^2$$

$$\theta_{t-1} = \theta_t - \eta \frac{\frac{m_t}{1 - \beta_1^t}}{\sqrt{\frac{v_t}{1 - \beta_2^t}} + \epsilon}$$

e. Model Evaluation

- **MAE:** $\frac{1}{n} \sum |y_i - \hat{y}_i|$
- **MSE:** $\frac{1}{n} \sum (y_i - \hat{y}_i)^2$
- **RMSE:** $\sqrt{MSE}$
- **R²:** $1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

3. Validation Methods

a. Holdout Validation

Split data into training and testing (e.g., 80/20):

$$\text{Train Set: } (X_{\text{train}}, Y_{\text{train}}), \quad \text{Test Set: } (X_{\text{train}}, Y_{\text{train}})$$

Evaluates generalization on unseen data.

b. Leave-One-Out Cross Validation (LOOCV)

$$RMSE_{LOOCV} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \widehat{y}_{(-i)})^2}$$

Every sample acts once as validation; gives unbiased performance estimate.

4. Prediction Inference

a. Forward Propagation

$$\hat{y} = f(X) = W_2 \cdot \sigma(W_1 X + b_1) + b_2$$

Generates predicted item performance probabilities.

b. Normalization

$$y' = \frac{y - \min(y)}{\max(y) - \min(y)}$$

Rescales predicted probabilities to [0,1] for IRT compatibility.

5. Item Response Theory (IRT) - Rasch Family

a. Partial Credit Model (PCM)

$$P(X_{ij} = k | \theta_i) = \frac{e^{\sum_{m=0}^k (\theta_i - b_{jm})}}{\sum_{h=0}^{M_j} e^{\sum_{m=0}^h (\theta_i - b_{jm})}}$$

- θ_i : person ability
- b_j : item difficulty
- $P(X_{ij} = 1)$: probability of correct/positive response.

Handles polytomous items (scores with multiple categories).

b. Rating Scale Model (RSM)

$$P(X_{ij} = k) = \frac{e^{k(\theta_i - b_j - \tau_k)}}{\sum_{m=0}^M m(\theta_i - b_j - \tau_m)}$$

Same thresholds τ_k across all items.

c. Ability & Difficulty Estimation (Joint Maximum Likelihood)

Iteratively estimate:

$$\theta_i^{(t+1)} = \theta_i^{(t)} + \frac{\sum_j (X_{ij} - P_{ij})}{\sum_j P_{ij}(1 - P_{ij})}$$

$$b_j^{(t+1)} = b_j^{(t)} - \frac{\sum_i (X_{ij} - P_{ij})}{\sum_i P_{ij}(1 - P_{ij})}$$

d. Fit Statistics

- **Infit:**

$$Infit_j = \frac{\sum_i w_{ij} z_{ij}^2}{\sum_i w_{ij}}, \quad w_{ij} = P_{ij}(1 - P_{ij})$$

- **Outfit:**

$$Outfit_j = \frac{\sum_i z_{ij}^2}{n_j}$$

- **Standardized t-statistics:**

$$t = \frac{z - 1}{SE(z)}, \quad p = 2(1 - \Phi(|t|))$$

f. Reliability (Person Separation Index)

$$Rel = \frac{Var(\theta)}{Var(\theta) + MeanSE_{\theta}^2}$$

Measures reproducibility of person ordering; bounded in $[0,1]$.

6. Wright Map (Person-Item Map)

Displays item difficulties (b) and person abilities (θ) on the same logit scale.

Axes:

- X-axis $\rightarrow$ logit scale (relative)
- Y-axis $\rightarrow$ count of persons/items
- Histograms show θ and b distributions.

Optionally overlays threshold step locations for PCM/RSM:

$$b_{jk}, \quad \tau_k$$

indicating category transition points.

7. Model Fit Diagnostics

a. Standardized Residual

$$z_{ij} = \frac{X_{ij} - P_{ij}}{\sqrt{P_{ij}(1-P_{ij})}}$$

b. Infit/Outfit Interpretation

Value	Interpretation
0.7–1.3	Acceptable fit
<0.7	Overfit (predictable)
>1.3	Underfit (unexpected)

8. Reliability and Variance Components

a. Person Variance

$$\sigma_{\theta}^2 = \text{Var}(\theta_i)$$

b. Error Variance

$$\sigma_{error}^2 = \text{Mean}(SE_{\theta_i}^2)$$

c. Reliability

$$Rel = \frac{\sigma_{\theta}^2}{\sigma_{\theta}^2 + \sigma_{error}^2}$$

9. Visualization Principles

- **Loss Curve:** visualize convergence (train vs validation loss).
- **Predicted vs Actual Scatter:** regression line $Y = X$.
- **Wright Map:** histograms + thresholds.
- **Progress Bars:** proportional epoch completion or item calibration.

10. Error and Stability Handling

- Replace all “.” and missing entries with “NaN” → filtered before numeric conversion.
- Clamp probabilities in $([10^{-6}, 1-10^{-6}])$ to avoid division overflow in IRT.
- Ensure θ and b re-centered:

$$\bar{\theta} = 0, \quad \bar{b} = 0$$

for interpretability.